

Pressure: Force per Unit Area

The same force feels very different over a sharp point or a broad strap — that difference is pressure.

What pressure means

Pressure is the force acting per unit area, on the surface perpendicular to the force.

$$\text{Pressure} = \text{Force} \div \text{Area} \quad (P = F / A)$$

SI unit: newton per square metre (N/m²), also called the **pascal (Pa)**. So 1 Pa = 1 N/m².

Same force, different pressure

- **Smaller area** → **more pressure**. A sharp nail tip or a knife's sharp edge concentrates force, so it cuts/pierces easily.
- **Larger area** → **less pressure**. Broad bag straps, a broad bucket handle, or a cloth pad under a head-load spread the weight, so they feel comfortable.

How to calculate

If a force of 100 N acts on an area of 2 m²: $P = 100 \div 2 = \mathbf{50 \text{ Pa}}$.

Watch units: force in N, area in m², pressure in Pa. Keep them consistent.

Where students lose marks

Trap 1 — Pressure depends on area, not just force. The same weight gives more pressure on a smaller area.

Trap 2 — Sharp = high pressure. A sharp edge/point has small area, so high pressure — that's why it cuts. Broad straps reduce pressure.

Trap 3 — Unit is pascal (Pa) = N/m². Don't confuse it with the newton (which is force).

Model answers — write them like this

Example A. A force of 60 N acts on an area of 3 m². Find the pressure.

Step 1: Use $P = F / A$.

Step 2: $P = 60 \div 3 = 20$.

$\therefore$ Pressure = 20 Pa (N/m²).

Example B. Why is it easier to cut with the sharp edge of a knife than the blunt edge?

Step 1: The sharp edge has a much smaller area of contact.

Step 2: For the same force, smaller area means greater pressure.

$\therefore$ Higher pressure under the sharp edge cuts more easily.

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